

Cybersecurity

Introduction to Digital Forensics

Lesson 21.2



Learning Objectives

By the end of today's lesson, you will be able to:

- 1 Summarize the basic principles and methodologies of digital forensics.
- 2 Describe various skill sets required for digital forensics jobs.
- 3 Outline the approach to collecting, preserving, analyzing, and reporting digital forensic evidence.
- 4 Conduct a preliminary review for a digital forensics case.
- 5 Preserve and document evidence using Autopsy.



Welcome



Overview

Overview

Digital forensics requires a unique skill set.

We learned in earlier modules that penetration tests attempt to discover vulnerabilities in order to prevent breaches and cybercrimes.

We know that not all defenses can withstand the constant network attacks of today's cyber landscape.

Digital forensics attempts to discover evidence of a crime **after** it has been committed.

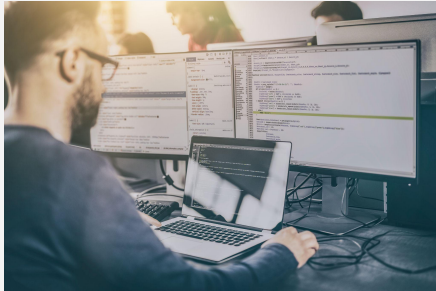


Roles, Knowledge, and Skill Sets

Digital forensics professionals must have a thorough understanding of:

01

Hardware



02

Operating Systems



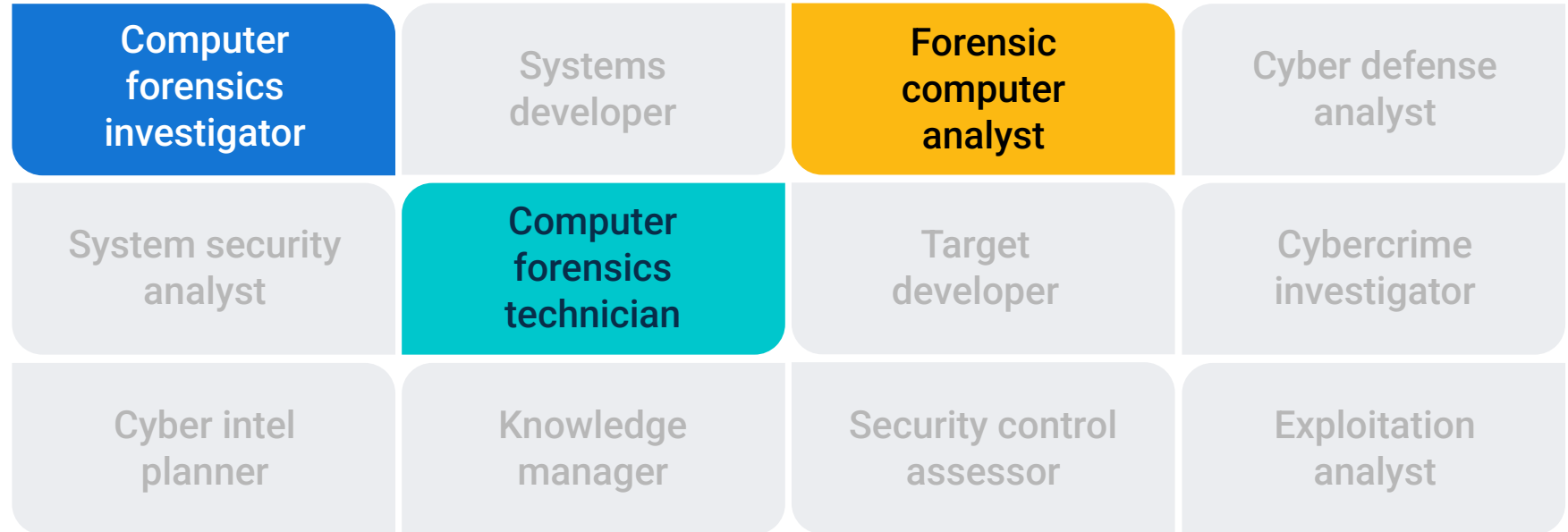
03

Computer Networks



Roles, Knowledge, and Skill Sets

The roles and responsibilities in the field of digital forensics are not well-defined across job titles. Specific job roles and responsibilities vary by organization.



Roles, Knowledge, and Skill Sets

Computer forensics investigator

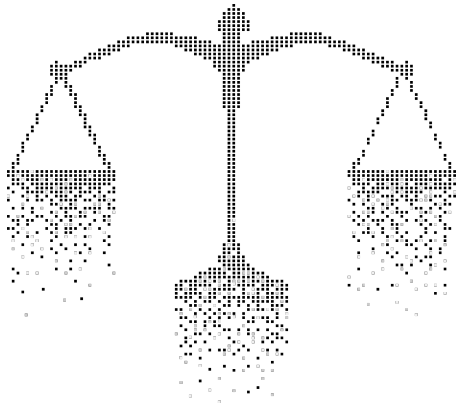
Job description

Gathers digital information for computer system investigations producing criminal evidence that can be used in a court of law.

Responsibilities

Legal cases involve data theft, espionage, and cyberwarfare.

This career field involves analyzing data found on hard drives, network communications, and cloud-based data communication systems, as well as the production of reports and possibly testifying in a court of law.



Roles, Knowledge, and Skill Sets

Computer forensics technician

Job description

Digital detective who works with law enforcement or government entities, or as a private investigator.



Responsibilities

Uses investigative and computer analysis techniques to acquire, analyze, and preserve digitized evidence to be used for legal purposes.

Inspects storage media and retrieves corrupted and deleted files from computing storage media, such as hard drives, removable flash media, and mobile devices. This material is used in legal proceedings.

Roles, Knowledge, and Skill Sets

Forensic computer analyst

Job description

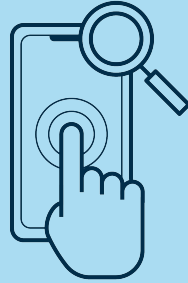
Specializes in the recovery of deleted emails or other data that has been encrypted or deleted. This material is used in legal cases involving computer crimes.

Responsibilities

Evidence produced by forensic computer analysts is used by law enforcement to assist with ongoing investigations.

Evidence may also be used in a court of law to help convict criminals involved in computer crimes with cases related to child pornography, drug sales, or corporate espionage, among others.





Introduction to **Digital Forensics**



What is **Digital Forensics**?



Digital forensics is a field dedicated to identifying, extracting, preserving, and reporting on information obtained from computer and network systems.





According to the article “Digital Records Forensics: A New Science and Academic Program for Forensic Readiness,” **digital forensics** can be defined as:

The use of scientifically derived and proven methods toward the preservation, collection, validation, identification, analysis, interpretation, documentation, and presentation of digital evidence derived from digital sources for the purpose of facilitating or furthering the reconstruction of events found to be criminal, or helping to anticipate unauthorized actions shown to be disruptive to planned operations.”

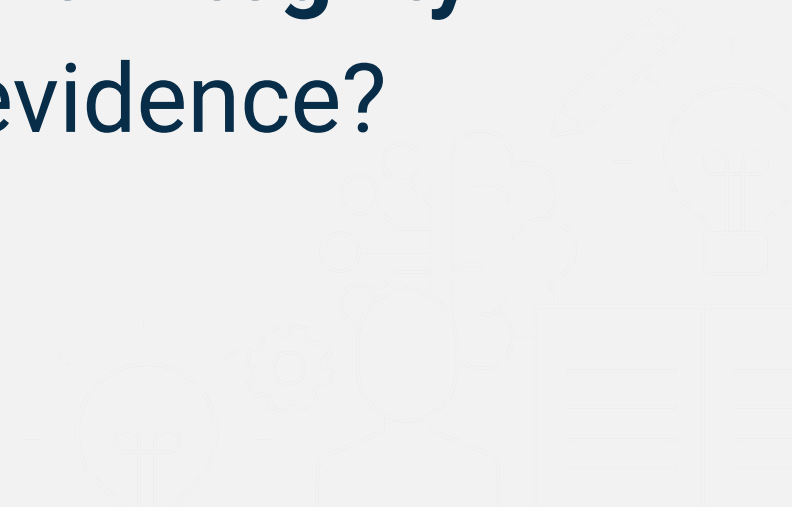
— *Journal of Digital
Forensics, Security and Law*



Legal Requirements



Why is it important to
preserve the **integrity**
of digital evidence?



Legal Requirements

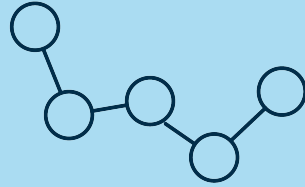
Digital forensics evidence that is intended for use in legal proceedings must satisfy a set of legal standards in order to be admissible in court.



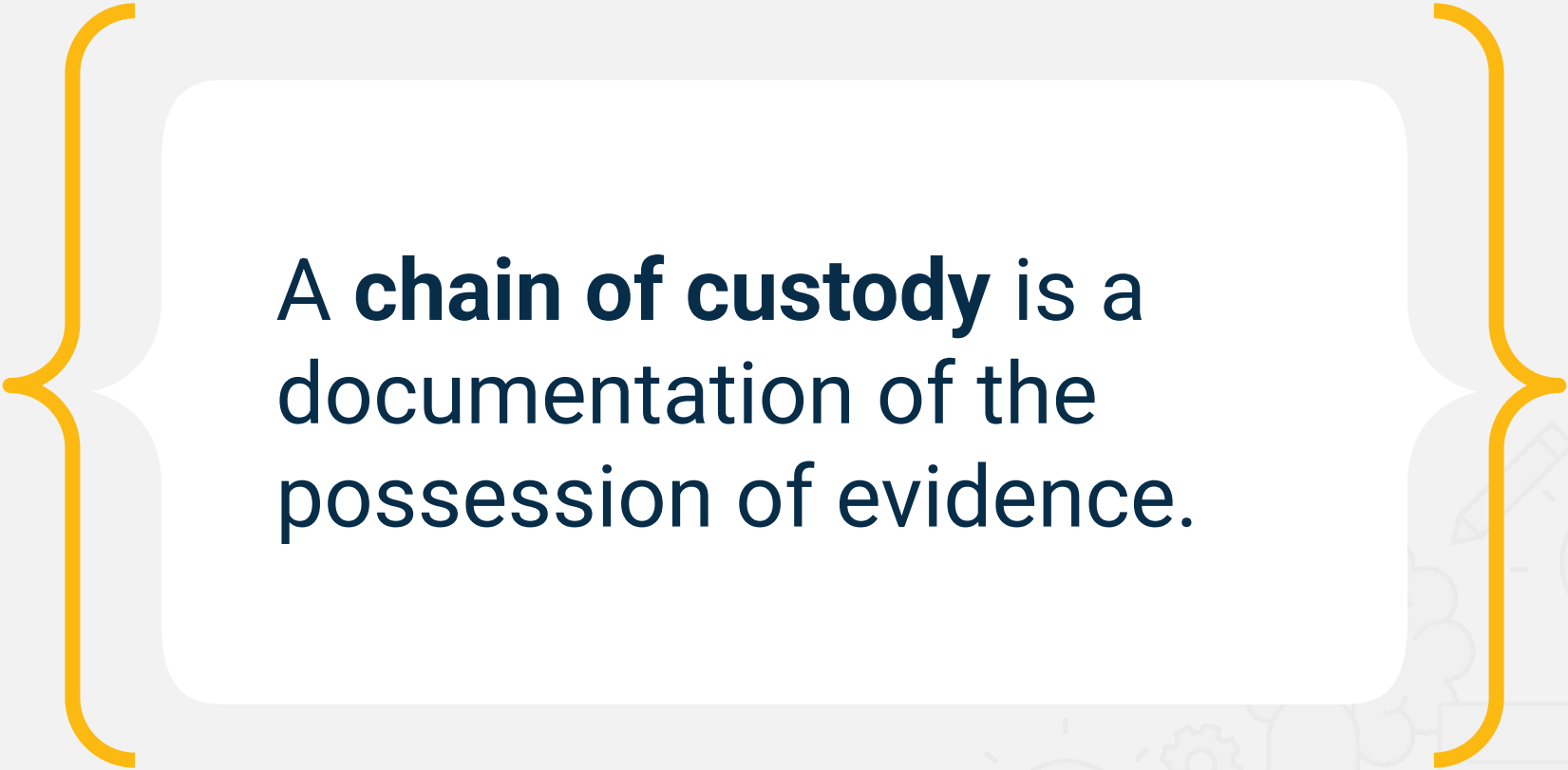
Imagine the disappointment of spending months searching through computer and network systems collecting evidence only to have it considered inadmissible and thrown out due to improper procedures.

Digital forensics evidence is held to the same standards as any other evidence submitted in a legal case. All evidence must be wholly intact and unaltered from the scene of the crime.

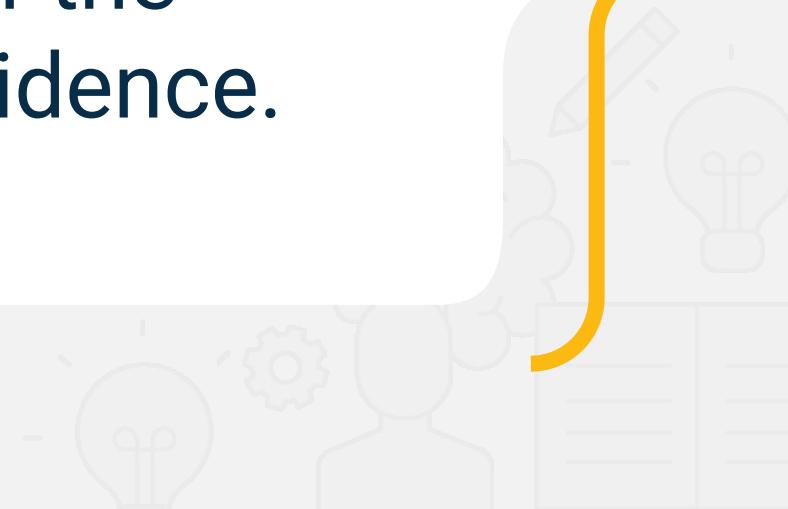
Once evidence is collected, the investigator or analyst is accountable for its ownership and whereabouts at all times.



Chain of **Custody**



A chain of custody is a documentation of the possession of evidence.



Chain of Custody

Evidence taken into custody must always be accounted for.

This means that its whereabouts and ownership must be accurately documented at all times, from the time it is acquired to when it is presented in the courtroom.



Chain of Custody

When testifying about the integrity of evidence, you must be able to provide a documented, uninterrupted chain of custody that includes:

01

A detailed log of every person who accessed or handled the evidence.

02

The purpose of that person's access.

03

The dates and times the evidence was handled.



If you are not able to show an uninterrupted chain of custody, months or even years of hard work can be considered inadmissible in a court of law.



The Digital Forensics Process

The Digital Forensics Process

The National Institute of Standards and Technology (NIST) defines a process for performing digital forensics in *Special Publication 800-86* as follows:

Collection	We must first collect the data before we can examine and analyze it. The collection phase is the springboard to the digital forensics process, which includes identifying, labeling, recording, and acquiring data from sources while following procedures to preserve the integrity of the data.
Examination	
Analysis	
Reporting	

The Digital Forensics Process

The National Institute of Standards and Technology defines a process for performing digital forensics in *Special Publication 800-86* as follows:

Collection	
Examination	After we've collected the evidence, we begin the examination phase, which ensures that all data collected is relevant to the case. This includes forensically processing collected data and assessing and extracting data of interest while preserving the integrity of the data. This usually means working from a copy rather than the original.
Analysis	
Reporting	

The Digital Forensics Process

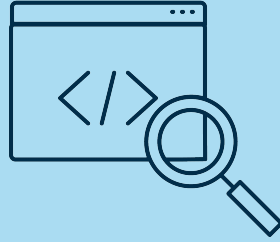
The National Institute of Standards and Technology defines a process for performing digital forensics in *Special Publication 800-86* as follows:

Collection	
Examination	
Analysis	After we've identified all relevant evidence, we begin the analysis phase by analyzing the results of the examination, using legally approved methods and techniques to derive information that addresses the questions that inspired the collection and examination.
Reporting	

The Digital Forensics Process

The National Institute of Standards and Technology defines a process for performing digital forensics in *Special Publication 800-86* as follows:

Collection	
Examination	
Analysis	
Reporting	After we've analyzed all of the evidence, digital forensics investigators are required to formally report the results of the analysis, which may include describing the actions used, explaining how tools and procedures were selected, determining what other actions need to be performed, and recommending improvements to the forensics process.



Evidence Acquisition

Evidence Acquisition

Evidence acquisition is one of the most critical steps in the entire forensics process.

If done improperly, it can result in evidence that is missed, lost, or inadmissible in a court of law.



Evidence Acquisition

Digital forensics data has one of two classifications:

Network-based

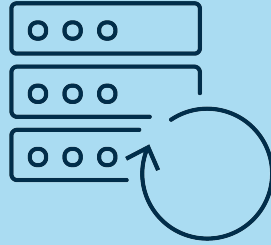
Comes from data communications captured by network-based systems such as an IDS, IPS, or firewall in the form of a packet capture or similar.

- Packet captures are useful for reconstructing events involving computer break-ins.
- Logs from firewalls also provide insight into network activity.

Host-based

Typically found on a system that encompasses a wide variety of artifacts.

- Investigations involving computer break-ins may involve the forensic examination of local file systems, programs, access to critical documents, and alterations to system files and directories.
- Host-based examinations may involve reconstructing internet use, unauthorized activity, email recovery, and the identification of malicious files.



Creating **Forensic Backups**

Creating Forensic Backups

During an investigation, it is important that evidence is not contaminated.

As a digital forensics investigator, you will **image** the data.

EVIDENCE

CASE NO. _____

SUSPECT _____ VICTIM _____

REMARKS _____

CHAIN OF CUSTODY

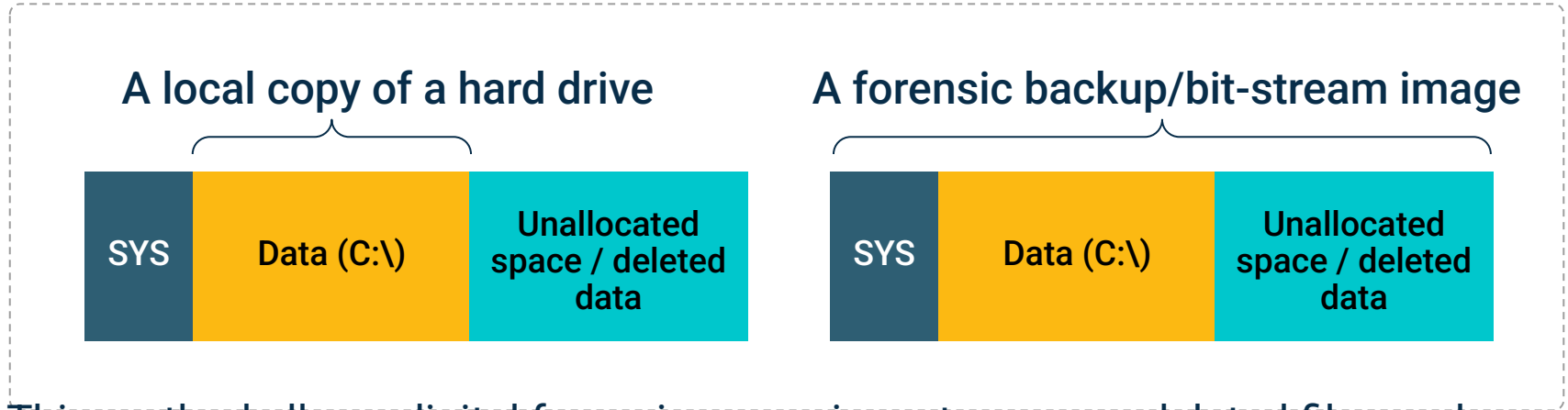
RECEIVED FROM	DATE	BY	TIME	AM	PM



This **image**, not the original media, will be used for analysis.

Creating Forensic Backups

A **bit-stream image** captures all created sections of a hard drive, known as partitions, whether used or not, and all unallocated drive space that doesn't belong to partitions.

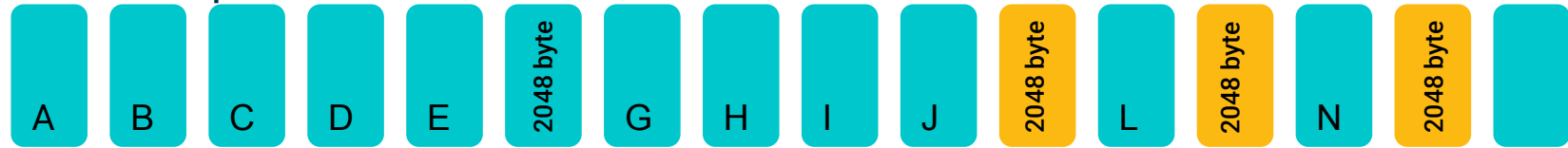


This method allows digital forensics examiners to recover deleted files and fragments of data that may exist on the hard drive.

Creating Forensic Backups

A **local file system** copy is inadequate for forensic analysis. If you make a copy through the file or operating system level, you can access only the data that the operating system can access. This won't capture deleted files or slack space. Therefore, you need to obtain a **bit-level copy**.

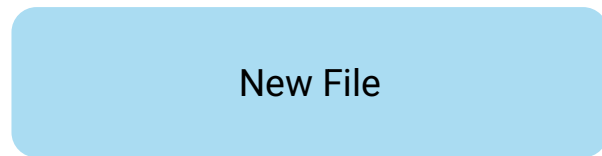
File: example.doc



Deleted file: delexample.doc



Overwritten file



Slack space



Forensic Disk Image Formats

Forensic backups (system image or bit-stream image) create an exact replica of all contents on the hard drive, including slack space, free space, and deleted files.

Forensic backup images are created in the following formats:

Raw format

Can be created with programs like **dd**, **ddfldd**, and **ddcdd**.

Examples:

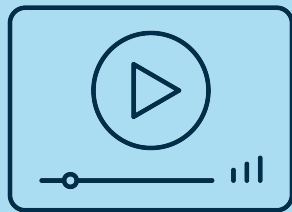
.bin
.dd
.img
.raw

Advanced Forensic Format (AFF)

For disk image and related forensic metadata

Examples:

.AFF
.AFF4



Working With **Live Systems**

You should always use caution when working with live systems.

It is possible that an attacker is waiting for the user to log back in to the system to complete the attack.

[!]

EXPLOIT



Working With Live Systems

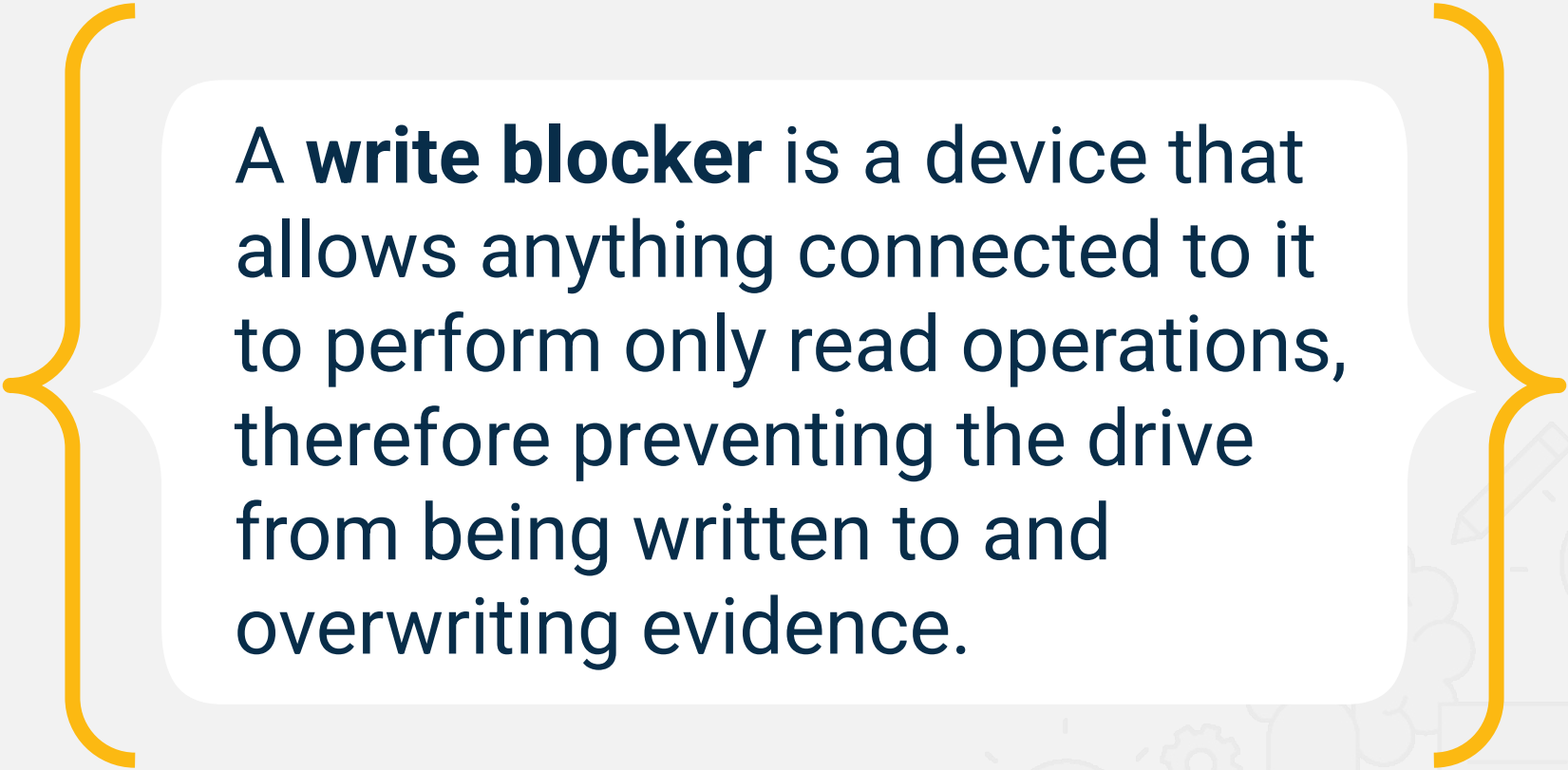
The primary reason for working on a live system is to capture items that will not survive a power loss, such as:

- 1 Volatile memory
- 2 Swap files
- 3 Running processes
- 4 Active network connections



Mistakes such as writing data to memory or disk can potentially destroy evidence, which is why it is critical to use digital forensics tools such as write blockers.





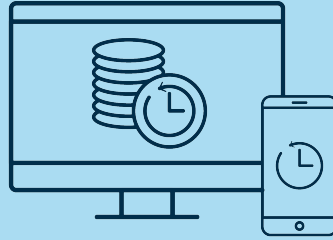
A **write blocker** is a device that allows anything connected to it to perform only read operations, therefore preventing the drive from being written to and overwriting evidence.



File Systems

During an investigation, you might encounter these file systems:

New Technology File System (NTFS):	Supported under Windows 10, 8, 7, Vista, XP, and NT
File Allocation Table (FAT):	Supported by older and newer versions of Windows
Apple File System (AFS):	Used by Mac OS systems
Fourth Extended File System (Ext4):	Used in most Linux distributions



Overview of **Storage Media**



What type of **storage devices** do you have on your computer, tablet, or mobile phone?

Storage Media Overview



**Mechanical
hard drives**



**Flash
storage**



**Solid-state
drives (SSD)**



**SD or microSD
cards**

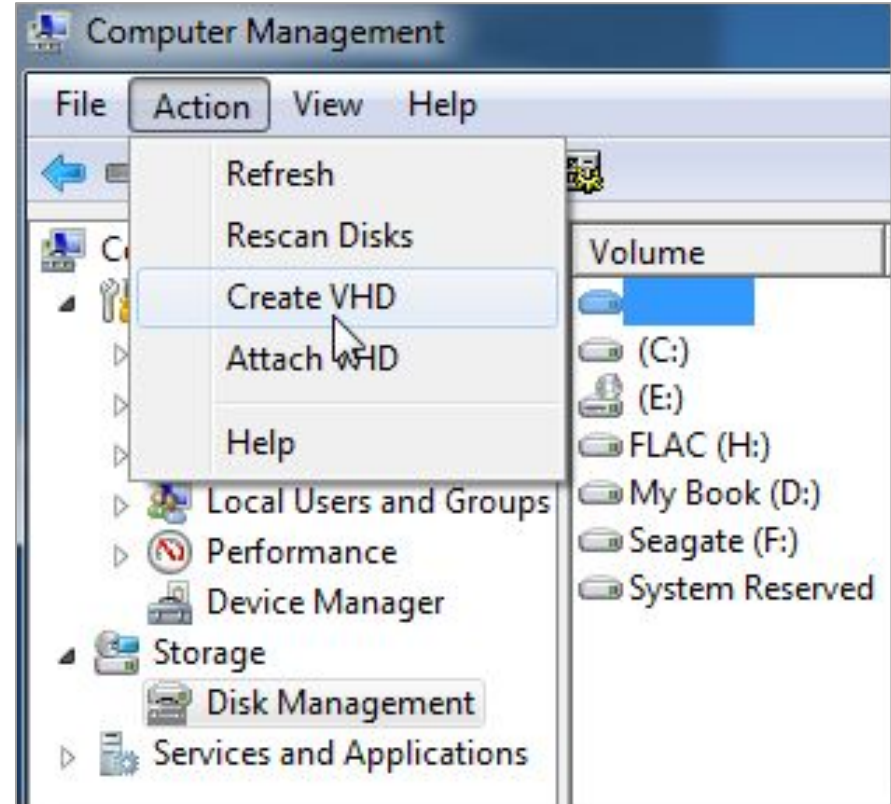
Physical vs. Virtual Drives

A **virtual drive** resides on a physical drive and emulates the characteristics of a hard drive.

The drive can be mounted and dismounted, and data can be read and written to it.



Drives in general are limited by the speed at which they can receive and send information.



Mechanical Hard Drives Have...

... delicate moving parts that can be damaged if not handled properly.



... larger storage capacities than other types of drives. This means the imaging process can take a long time when performing a bit-level copy.



Implications for Forensics

Data can be recovered from badly damaged or comprised devices. This requires extensive knowledge of the inner workings of hard drives and how hard drives store data.

Flash Storage

Flash storage devices have no moving parts and use flash memory, allowing for quicker read and write access. Flash memory or flash storage is non-volatile, meaning it holds data even when it's not connected to power.

This technology is used in devices such as:



USB drives



Mobile phones



Cameras



Tablet computers

Solid-State Drives (SSD)

SSD storage devices use flash memory chips and have no moving parts. They also have larger storage capacity and faster read and write access.



Implications for Forensics

A SSD is not a mechanical drive. You must be careful with the forensics tools used to image and recover data. The data on a SSD can be lost or wiped out **in seconds**.

SD or MicroSD Cards

SD cards store data inside a flash memory chip, similar to solid-state devices.

They are used in cell phones and smartphones.



Implications for Forensics

It is possible to retrieve data from a SD card even if it has been reformatted or the data has been deleted. This is because data is not immediately erased; the storage is only set aside for reuse. As long as the storage space has not been used, the preexisting data is still available. This is very useful for digital forensics recovery.

Firmware

You may encounter password-protected storage media and password-protected **firmware**. Firmware is a specific class of computer software that provides the low-level control for a device's specific hardware.



Such situations present extreme challenges to digital forensics investigators and must be considered during the initial stages of the investigation.

Working with Cloud-Based Evidence

Working with online evidence, such as hard drives and volatile memory located on cloud-based hardware, presents a whole new set of challenges.





Can you think of any
challenges associated with
cloud-based evidence?



Questions?





Activity

Digital forensics in the cloud

In this activity, you will look at a case concerning a DDoS attack that occurred on an online shopping website.

- You will list at least four challenges associated with conducting a cloud-based forensics investigation.
- You will also research how these challenges affect the chain of custody during a digital forensics investigation.

Suggested Time
20 minutes



Time's Up!
Let's review



Questions?





Digital Forensics **Types** **and Methodology**

Digital Forensics Types

In addition to cloud-based forensics, digital forensics is a continuously evolving scientific field that incorporates many subdisciplines:

- 1 Computer forensics
- 2 Disk forensics
- 3 Memory forensics
- 4 Network forensics
- 5 Email forensics
- 6 Mobile device forensics





Digital Forensics Types

Computer forensics

Computer forensics is the identification, preservation, collection, analysis, and reporting of evidence acquired from computers, notebooks, and storage media. It is used to support investigations and legal proceedings.

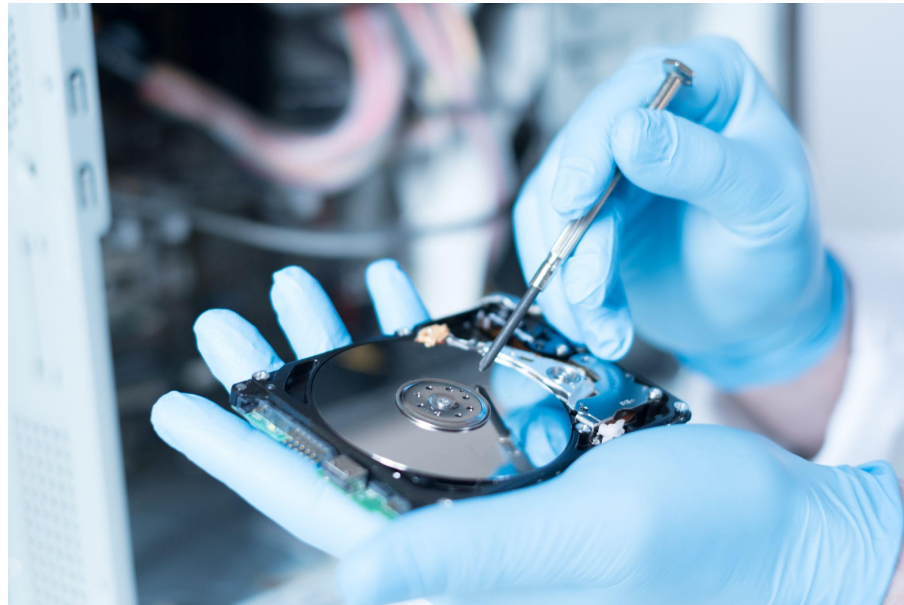




Digital Forensics Types

Disk forensics

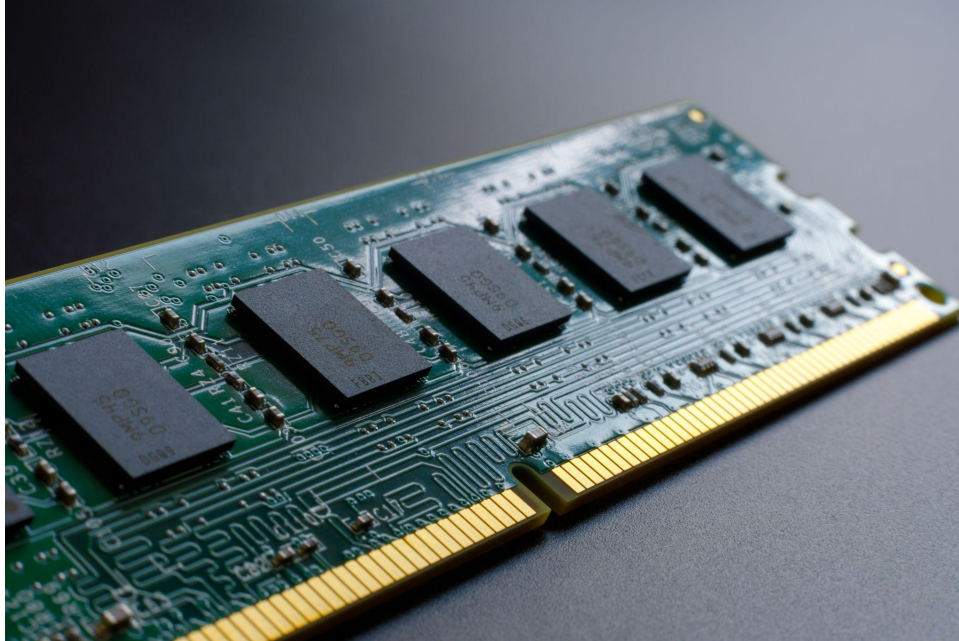
Disk forensics involves acquiring and analyzing information stored on physical storage media, such as hard drives, smartphones, GPS systems, and removable media. This involves the recovery process of hidden data, deleted data, and slack space information.





Digital Forensics Types

Memory forensics



Memory forensics inspects computer memory to identify an attacker's activities on a system.



Digital Forensics Types

Memory forensics requires a broad skillset

- You must have knowledge of CPU architectures, operating systems, memory management, page tables, and virtual addressing.
- Memory forensics is different from disk forensics in that the data investigated is stored in volatile memory, such as RAM.
- Data on hard drives does not change between reboots, so it is non-volatile.



Digital Forensics Types

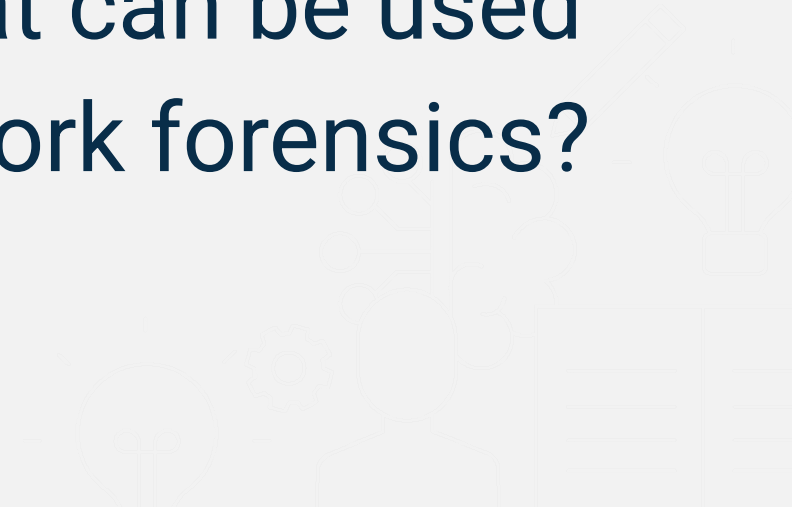
Network forensics



Network forensics involves the monitoring, recording, storing, and analysis of all network traffic to determine the source of security events.



Can you think of any
tools that can be used
for network forensics?





Digital Forensics Types

Network forensics tools include Wireshark, NetworkMiner, and Snort

01

Wireshark



02

NetworkMiner



03

Snort





Digital Forensics Types

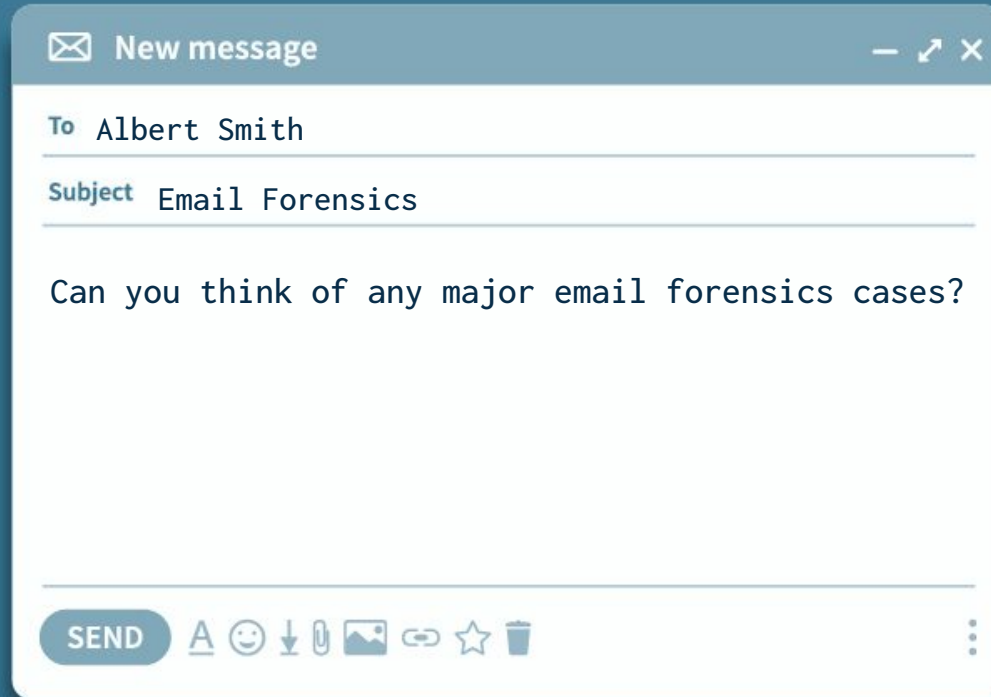
Email forensics

Email forensics analyzes the source and content of an email as evidence.



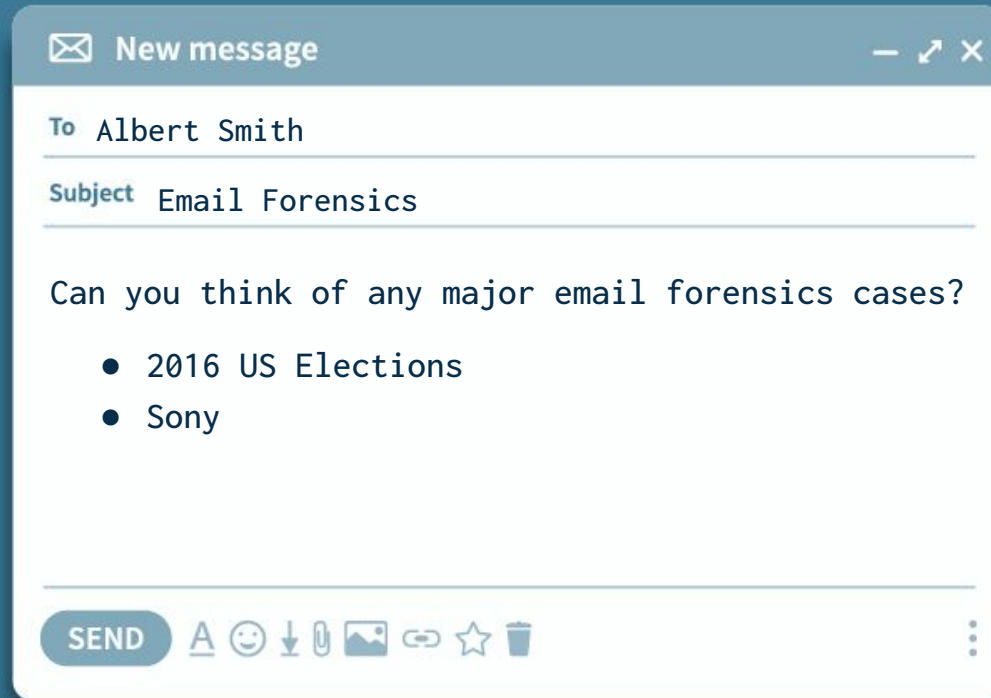
Email Forensics

Email forensics includes the process of identifying the sender, recipient, date, time, and original location of an email.



Email Forensics

Email forensics includes the process of identifying the sender, recipient, date, time, and original location of an email.





Digital Forensics Types

Mobile device forensics

Mobile device forensics is the recovery of digital evidence from smartphones, GPS devices, SIM cards, personal digital assistants, tablets, and game consoles.

For example:

Mobile device forensics might be used in a distracted driving case. A digital forensics expert could analyze what was happening at the time of the accident.





Digital Forensics Types

Other forensics areas include:

01

Drone

02

Internet

03

Malware technologies



Questions?





Digital Forensics **Investigation Methodology**

Digital Forensics Investigation Methodology

A successful digital forensics investigation is crucial for the prosecution of computer criminals.

Having a clearly defined step-by-step process for collecting digital evidence allows it to be permissible in a court of law.



Methodology for Conducting an Investigation

Digital forensics has a number of investigative frameworks. They include variations of the following phases:

01

Prepare for the investigation

02

Collect the evidence / digital forensics recovery

03

Preserve the evidence

04

Electronic discovery and analysis

05

Present and report

Step 1: Prepare for the Investigation

Preparation is an important first step in conducting an investigation. However, there is not always a lot of time for it.

As an investigator, you want to consider such things as:

01

Whether the incident was remote or local

02

What laws are relevant

03

What tools should be used (GUI or CLI)

Step 2: Collect the Evidence / Digital Forensics Recovery

The success of the analysis phase depends on the collection phase.

01

During the collection phase, the investigator makes decisions about what data to collect and the best way to collect it.

02

During digital forensics recovery, the evidence is extracted from a device and a master copy is made.

03

How you collect the evidence determines if the evidence is admissible in court.

Step 3: Preserve the Evidence

Investigators never work on the original copy of the evidence. A read-only master is made and stored in a digital vault.

01

All processing is done on the copy.

02

Investigators use a write blocker to ensure the prevention of data contamination.

Step 4: Electronic Discovery and Analysis

Analysis is done after data has been collected. This is sometimes referred to as dead analysis. It's important to document everything, including:

01

Time

02

Date

03

Locations

04

Applications used

Step 5: Present and Report

In this phase, investigators write an expert report that lists:

01

All tests conducted

02

When and how they were conducted

03

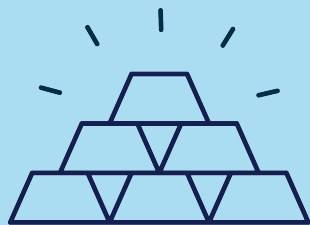
What was found

04

The conclusions of the investigation



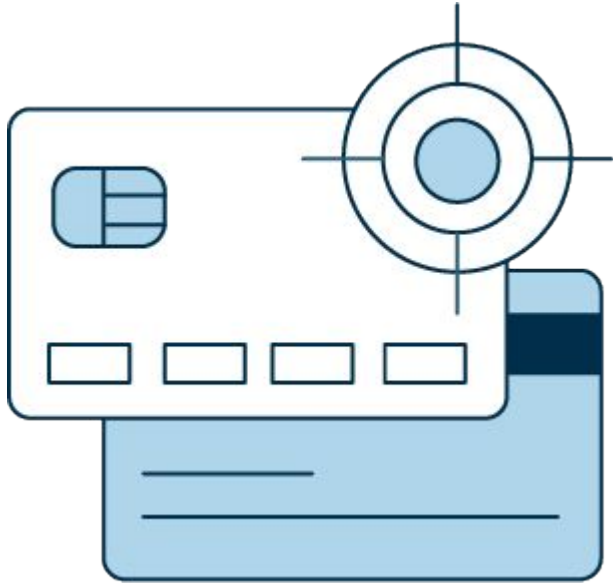
Digital forensics investigators may be called to testify as expert witnesses in trials or depositions.



Pure Gold Credit Union Case



Class Scenario



Over the next two lessons, you will work on a case involving a fictitious bank theft at **Pure Gold Credit Union**.

- The scenario is centered around employees at Pure Gold Credit Union who conspired to steal funds from their credit union branch.
- Suspicious activity was reported to senior management and devices were seized, including physical images of mobile devices.
- The seized evidence was processed by the ingest team at the crime laboratory.
- The evidence was backed up using the EnCase digital forensics application.



Activity

The Pure Gold Credit Union case

In this activity, you will work as a digital forensics investigator conducting an investigation into the Pure Gold Credit Union Case.

Suggested Time
15 minutes





Time's Up!
Let's review

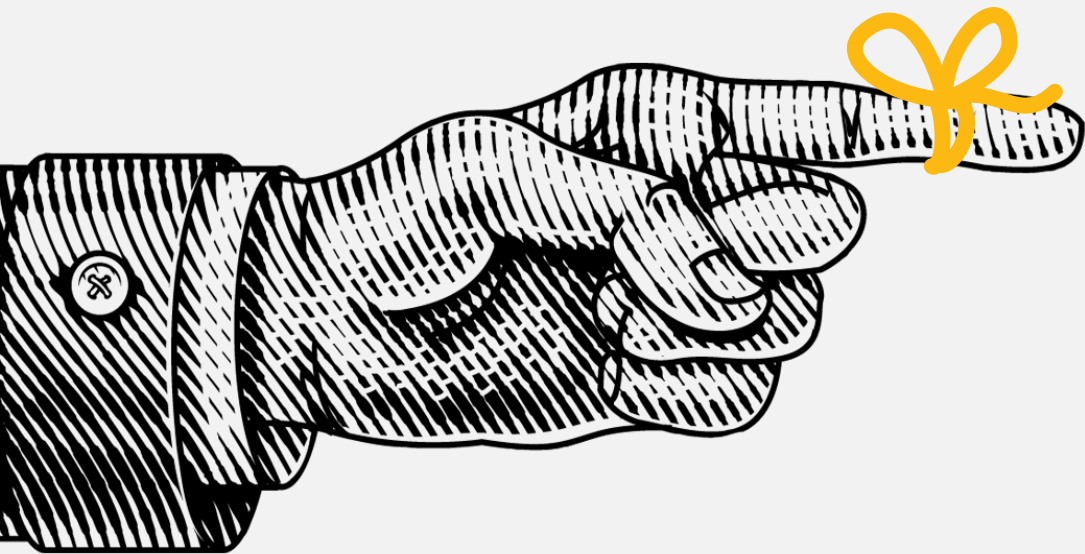


Questions?





Introduction to **Autopsy**



Remember

The seized evidence included an iPhone image.

iPhones have storage devices that hold:

- Operating system data
- File system data
- Applications
- Documents
- Pictures
- Videos
- Music

Introduction to Autopsy

We'll be looking for evidence related to the case, including:

01

SMS messages that show intent to commit a crime.

02

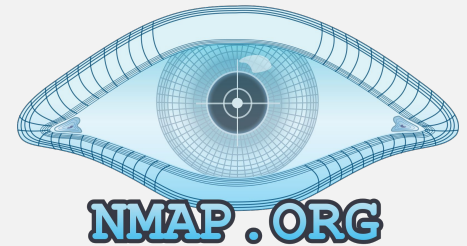
IP address information that indicates GPS location data.

03

Emails and email addresses that contain incriminating evidence.

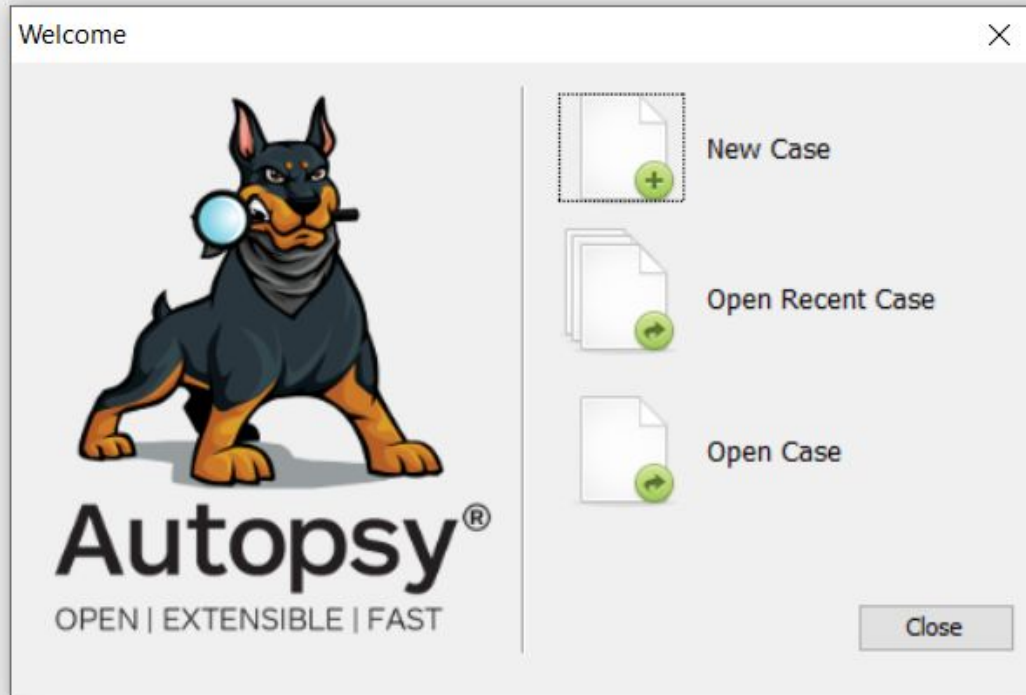
Open Source Tools

Digital forensics relies on the expertise of examiners to analyze and interpret data retrieved by trusted forensics examination tools. Although most digital forensics examiners have started to use open source tools, they may not know that they can perform a complete investigation using only open source tools.



Autopsy | Digital Forensics

We will use **Autopsy**, an open source digital forensics platform and graphical interface to The Sleuth Kit that runs on Windows, Ubuntu, Kali Linux, and OS X.



Autopsy: Prepare the Data

First we'll prepare the data by:

01

Running a virus scan on the image.

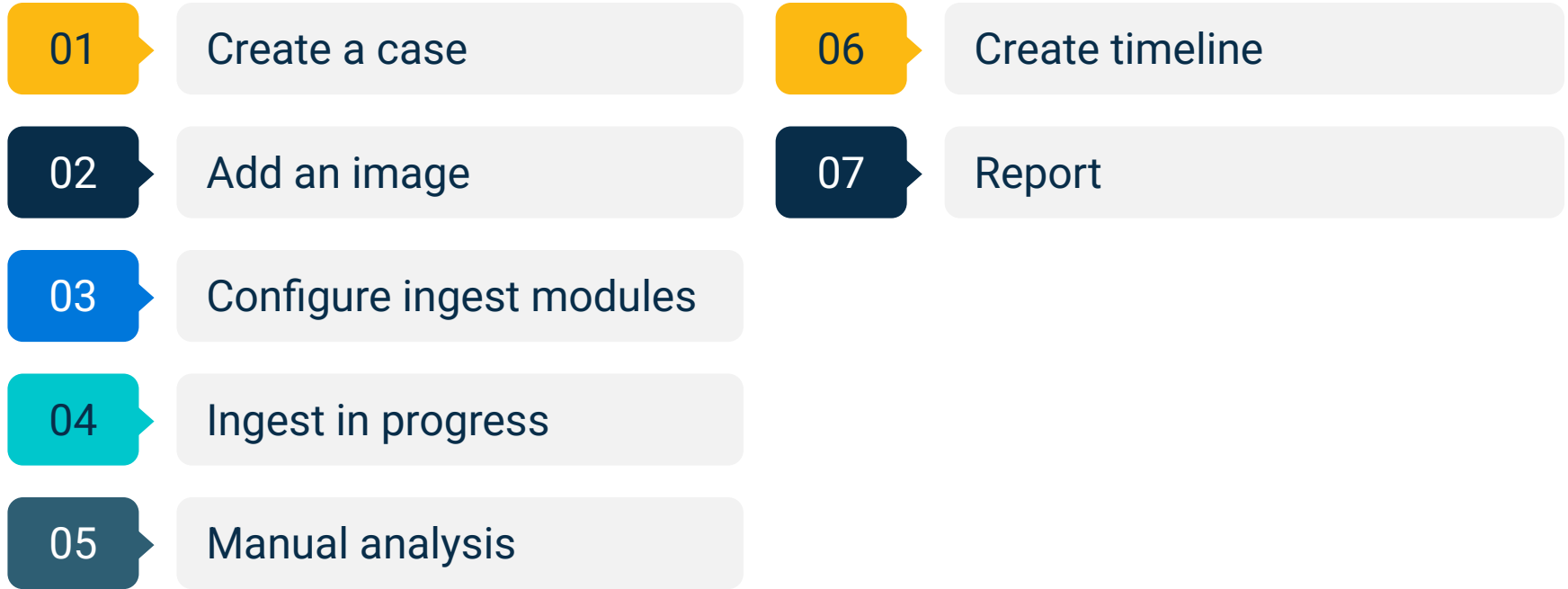
02

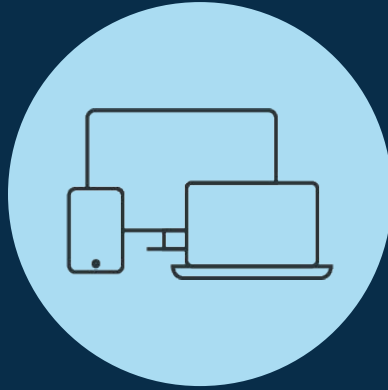
Generating an MD5 and SHA-256 hash for the evidence. This is to validate that nothing was changed during the investigation.

03

Opening a terminal window in Kali and navigating to
`/home/kali/Desktop/imagefiles`

The Autopsy Workflow





Instructor Demonstration

Autopsy



Break
15 mins

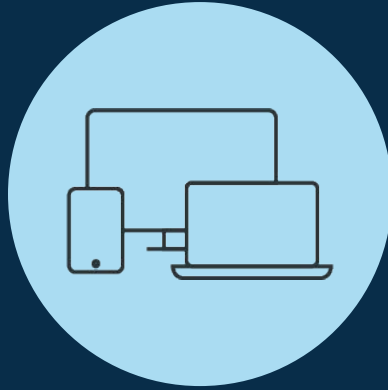
Now we'll pick back up at

Step 5

Manual Analysis with Keyword Search



Keyword Search



Instructor Demonstration

Autopsy continued



Let's Recap



Recap

Digital forensics

01

Although most digital forensics examiners have adopted the use of open source tools, many of them may not know that they can perform a complete investigation using only open source tools.

02

Autopsy is an open-source, multi-platform, graphical interface tool that runs on Windows, Ubuntu, Kali Linux, and OS X platforms.



Recap

Autopsy investigation steps

01

Generate MD5 and SHA-256 hashes for the evidence in order to ensure the integrity of the evidence.

02

Specify a case name and number to help investigators keep track of the files and their progress.

03

Select the file type to be ingested, i.e., .zip, and specify a working directory where progress will be saved.

04

Specify a standard time zone for the case.

05

Configure ingest modules to help categorize, label, and organize data.



Next Class

We'll continue our investigation



Thank You